

Faculty of Computing



Operating Systems

Lab # 5

Momna Khurram Qureshi

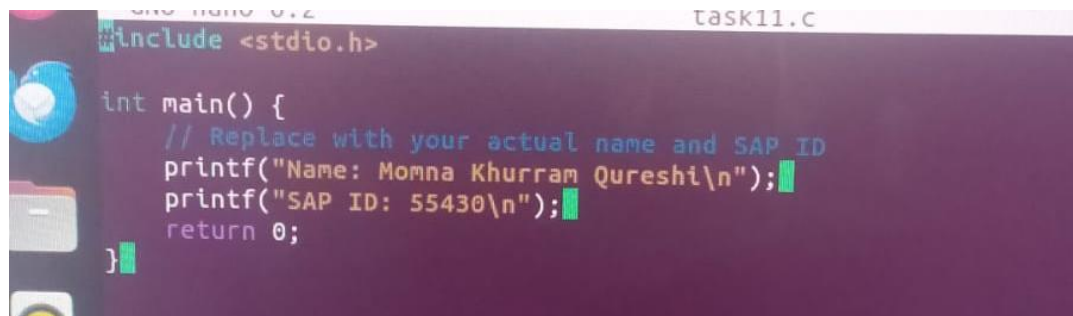
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Faculty of Computing,

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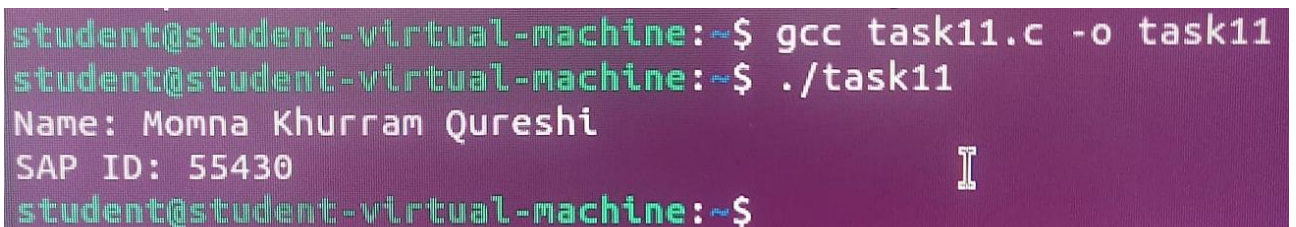
Task 1

Write a C program that print your name and sap id and Show the output.



```
#include <stdio.h>

int main() {
    // Replace with your actual name and SAP ID
    printf("Name: Momna Khurram Qureshi\n");
    printf("SAP ID: 55430\n");
    return 0;
}
```

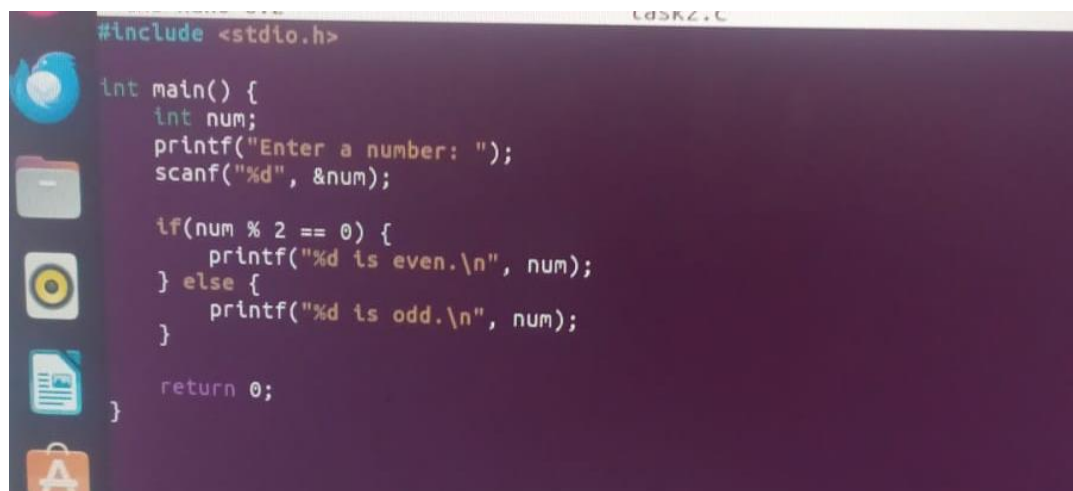


```
student@student-virtual-machine:~$ gcc task11.c -o task11
student@student-virtual-machine:~$ ./task11
Name: Momna Khurram Qureshi
SAP ID: 55430
student@student-virtual-machine:~$
```

Task 2

Write a C program that asks the user to enter a number and determine whether the number entered is even or odd.

Show the output.



```
#include <stdio.h>

int main() {
    int num;
    printf("Enter a number: ");
    scanf("%d", &num);

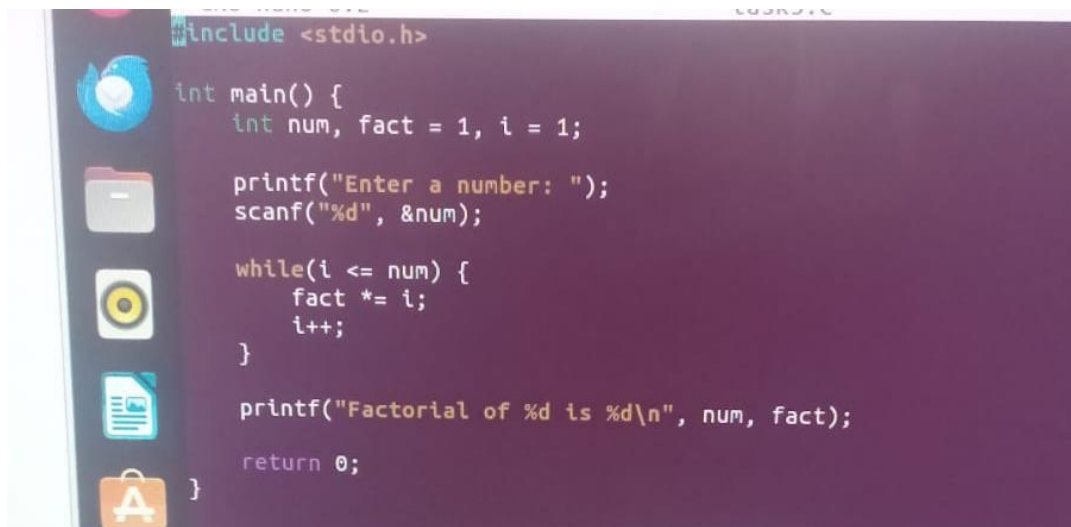
    if(num % 2 == 0) {
        printf("%d is even.\n", num);
    } else {
        printf("%d is odd.\n", num);
    }

    return 0;
}
```

```
student@student-virtual-machine:~$ pico task2.c
student@student-virtual-machine:~$ gcc task2.c -o task2
student@student-virtual-machine:~$ ./task2
Enter a number: 4
4 is even.
student@student-virtual-machine:~$
```

Task 3

Write a C program to calculate the factorial of a number using a while loop take a number from user.

A screenshot of a code editor with a dark theme. On the left side, there is a vertical sidebar with several icons: a blue circular icon with a white swirl, a folder icon, a yellow circular icon with a black center, a document icon, and an orange shopping bag icon. The main area of the editor displays the following C code:

```
#include <stdio.h>

int main() {
    int num, fact = 1, i = 1;

    printf("Enter a number: ");
    scanf("%d", &num);

    while(i <= num) {
        fact *= i;
        i++;
    }

    printf("Factorial of %d is %d\n", num, fact);

    return 0;
}
```

```
student@student-virtual-machine:~$ ./task3
Enter a number: 3
Factorial of 3 is 6
student@student-virtual-machine:~$
```

Task 4

Write a C program that uses an infinite for loop to print a name continuously.

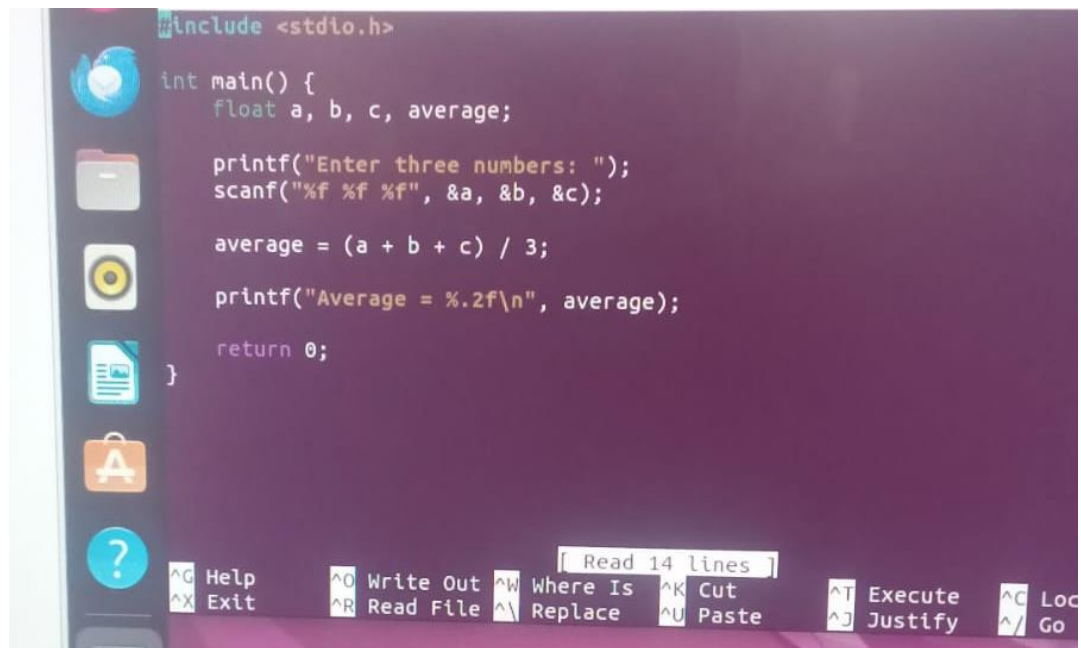
```
#include <stdio.h>

int main() {
    for(;;) {
        printf("Momna\n");
    }
    return 0;
}
```



Task 5

Write a program to print average of three numbers in c program.



A screenshot of a terminal window showing a C program. The program includes `<stdio.h>` and defines a `main` function. It declares three float variables `a`, `b`, and `c`, and a float variable `average`. It prompts the user to enter three numbers, reads them using `scanf`, calculates the average as `(a + b + c) / 3`, and prints the average using `printf` with a format of `%.2f`. The program returns 0. The terminal window has a sidebar with icons for a file manager, a terminal, a code editor, and a help icon. A status bar at the bottom shows various keyboard shortcuts.

```
#include <stdio.h>

int main() {
    float a, b, c, average;

    printf("Enter three numbers: ");
    scanf("%f %f %f", &a, &b, &c);

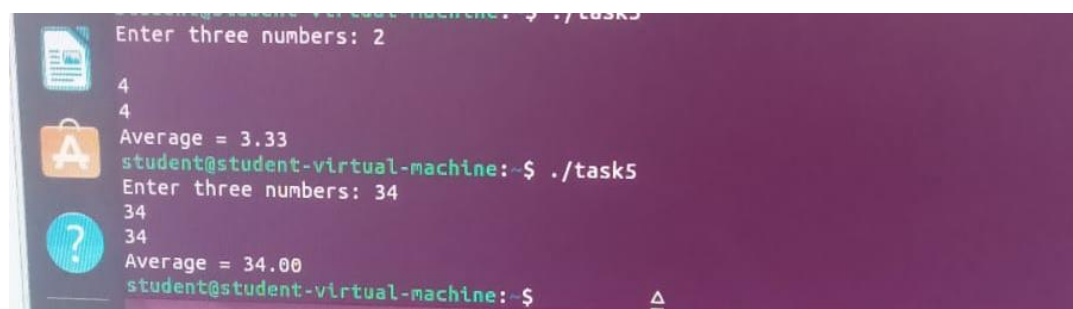
    average = (a + b + c) / 3;

    printf("Average = %.2f\n", average);

    return 0;
}
```

Read 14 lines

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Local Go



A screenshot of a terminal window showing the execution of the C program. The prompt is `student@student-virtual-machine:~$ ./tasks5`. The user enters three numbers: 2, 4, and 4. The program outputs `Average = 3.33`. The user then enters three numbers: 34, 34, and 34. The program outputs `Average = 34.00`. The prompt is `student@student-virtual-machine:~$`.

```
student@student-virtual-machine:~$ ./tasks5
Enter three numbers: 2
4
4
Average = 3.33
student@student-virtual-machine:~$ ./tasks5
Enter three numbers: 34
34
34
Average = 34.00
student@student-virtual-machine:~$
```